Reading the relationships learned by genomic LMs

Nucleotide dependency analysis of genomic language models detects functional elements

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Journal club notes

What is encoded in a genome language model?

A model can reconstruct a base accurately for very different reasons.

1. A repeated sequence is easy to copy.
2. A motif constrains a set of nearby bases.
3. An RNA fold links distant bases.

The paper's question
Can we expose these relationships directly, without a labeled downstream task?

Measure how a change at one base alters predictions elsewhere.

Introduction; Fig. 1a,b. gLM = genomic language model.

Core idea: perturb the input, observe the predictions



The query is the input position we change. The target j is the output position whose distribution we inspect. All other input bases stay fixed.

The published computation leaves every position unmasked, including the target.

Fig. 1b; Methods, Nucleotide dependencies and variant influence score.

Define a dependency before drawing a map

Let $k \in \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\}$ be a possible target base, and $a \neq x_i$ a query alternative.

$$O_j(k \mid x) = \frac{p_j(k \mid x)}{1 - p_j(k \mid x)}, \qquad \Delta_{ij}^{a}(k) = \log_2 \frac{O_j(k \mid x^{i \to a})}{O_j(k \mid x)}.$$

One specified variant

$$e_{ij,a} = \max_k |\Delta_{ij}^{a}(k)|.$$

Which target outcome changes most?

A query–target map entry

$$e_{ij} = \max_{a \neq x_i} \max_k |\Delta_{ij}^{a}(k)|.$$

Search all 3×4 perturbation/outcome pairs.

No self-dependencies ($i = j$); map rows are queries and columns are targets.

Methods. Notation rewritten; the maximum includes absolute log-odds changes.

Toy example: changing one side of a base pair

Suppose query i is T and target j is A in a putative RNA stem. Change query T $\rightarrow$ A.

Target prediction	Reference	Mutated query
$p_j(\mathrm{T})$	0.10	0.60
Odds of target T	$0.10/0.90 = 1/9$	$0.60/0.40 = 3/2$

$$\Delta_{ij}^{\mathrm{A}}(\mathrm{T}) = \log_2(13.5) \approx 3.75.$$

This one outcome contributes 3.75 to the maximum; other target bases and query alternatives must also be checked.

Interpretation: the model shifts toward the complementary target T. The assay measures a change in model prediction, not an observed compensatory mutation.

Illustrative probabilities invented for teaching; biological analogy: Fig. 1d.

Three summaries answer three different questions

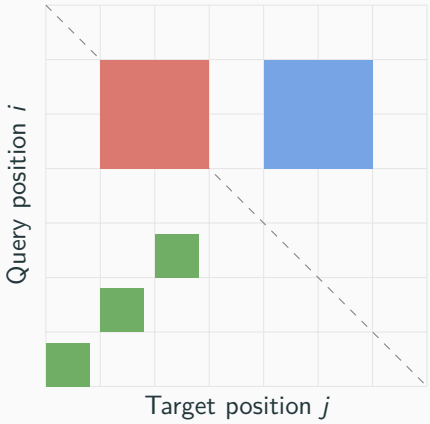
Quantity	Operation	Question
Reconstruction score	Log probability ratio at the variant position	Is this alternate base unlikely?
Variant influence	$I_{i,a} = \frac{1}{N-1} \sum_{j \neq i} e_{ij,a}$	How much does this variant alter other predictions?
Contact score	$S_{ij} = \max(e_{ij}, e_{ji})$	How strongly are these two positions linked?

The raw map is directional: e_{ij} need not equal e_{ji}. Symmetrization is a later task-specific operation.

Influence scores average over a model's sequence window; they are not signed expression changes.

Methods, dependency scoring, variant impact benchmarks and RNA structure benchmarking.

How to read the geometry of a dependency map



Dense diagonal block: a local sequence motif.

Off-diagonal block: linked sequence elements.

Antiparallel diagonal: base pairing or inverted repetition.

Parallel diagonal: forward duplication.

Geometry suggests a hypothesis; context and controls determine its meaning.

Schematic, not measured data. Pattern interpretation: Figs. 2–5.

Which models supply the probabilities?

1. **SpeciesLM fungi / metazoa**: main demonstrations; trained on sequences 5′ of start codons. Compact fungal genomes make these windows include coding regions and ncRNAs as well as promoters.
2. **RiNALMo**: RNA contact analyses; pretrained on 36 million noncoding RNA sequences, both annotated and unannotated. Its pretrained LM is probed without contact fine-tuning.
3. **SpliceBERT**: human aberrant splicing examples; trained on vertebrate RNA sequences.

Fourteen gLMs were assessed overall. The framework is a probing procedure, not a single newly supervised predictor.

SpeciesLM metazoa uses 2-kb upstream regions from 494 metazoan species; tokenization differs across model families.

Methods, SpeciesLM training; Extended Data Table 1; Results, RNA structure. ncRNA = noncoding RNA.

Variant influence predicts measured promoter disruption

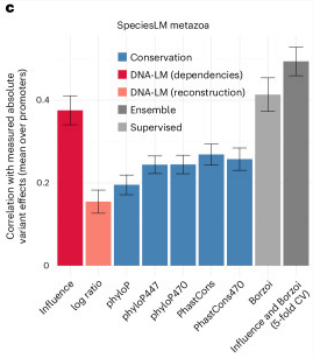


Fig. 1c benchmark

9 human promoters; 8,635 SNVs from saturation mutagenesis.

Influence correlates with absolute expression effects better than reconstruction and conservation.

Comparable to supervised Borzoi here; their combination improves prediction in fivefold CV.

This result concerns magnitude, not the direction of expression change.

Fig. 1c, cropped; Methods, promoter saturation mutagenesis and integrative model.

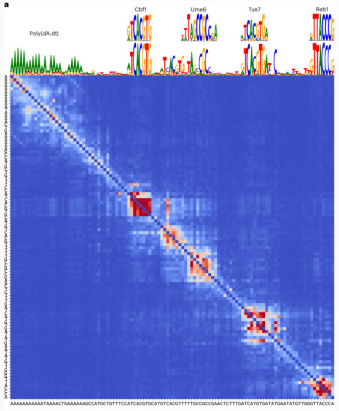
Variant benchmarks: useful signal, qualified superiority

Benchmark	Evidence	Qualification
ClinVar noncoding	22,313 pathogenic; 363,259 benign. Influence beats reconstruction.	Does not beat conservation; labels may incorporate conservation evidence.
Human promoter eQTLs	1,631 high-PIP versus 1,327 low-PIP variants.	Beats conservation and reconstruction, not Borzoi; PIP is an inferred label.
Yeast cis-eQTLs	379 MPRA-supported versus 2,677 controls.	Supports cross-species utility; selected loci and assays.

Promoter saturation mutagenesis provides the cleanest direct comparison with experimentally measured variant effects in this paper.

Extended Data Fig. 1a–f; Methods. SNV = single-nucleotide variant; PIP = posterior inclusion probability.

Dense blocks distinguish motifs from easy reconstruction



Yeast *SMT3* promoter: poly(dA:dT) and TF motifs are both reconstructed confidently.

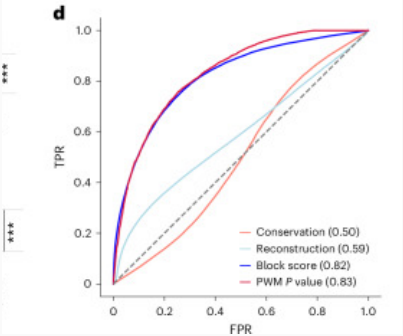
Motif bases strongly alter predictions of other motif bases, making compact blocks.

A single change within a long homopolymer is less disruptive to prediction elsewhere.

High probability alone does not establish functional importance.

Fig. 2a, cropped; TF = transcription factor. Crops preserve the original sequence and map.

A simple block score approaches expert motif scanning



Score dense dependencies using the first quartile within six-base windows.

Bound sites: PWM matches near ChIP-exo peaks, covering 68 yeast TFs; 92,117 positive bases among 6,538,427.

AUROC: block 0.82; PWM 0.83; reconstruction 0.59; conservation 0.50.

Labels use PWMs, favoring that comparator. Bound versus unbound PWM matches also carry block-score signal.

A fixed six-base score cannot capture every motif shape; Abf1 appears as spaced blocks.

Fig. 2d, cropped; Fig. 2e; Extended Data Table 2; Methods, TF motif mapping.

Distal blocks link regulatory sequence elements

Promoter architecture
TATA and initiator elements in a fly promoter show off-diagonal dependencies.

Yeast splicing
Donor, branch point and acceptor can depend on one another. True donor–acceptor pairs exceed internal AG decoys and distance-matched controls.

These controls address the strong baseline decay of dependencies with genomic distance.

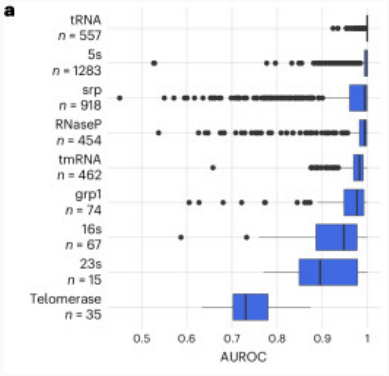
Human splicing outliers
SpliceBERT links a disrupted canonical donor in *TRPC6* with a cryptic splice site.

1,811 rare-variant-associated events from 946 GTEx individuals show stronger dependencies to outlier junctions than distance-matched backgrounds.

Association with altered junction use supports biological relevance; it does not identify a physical contact.

Fig. 3a–e; Extended Data Fig. 3. Events filtered to fit an 800-bp window within SpliceBERT context.

Antiparallel stripes recover RNA stems



Two paired runs go in opposite sequence order, creating an antiparallel diagonal.

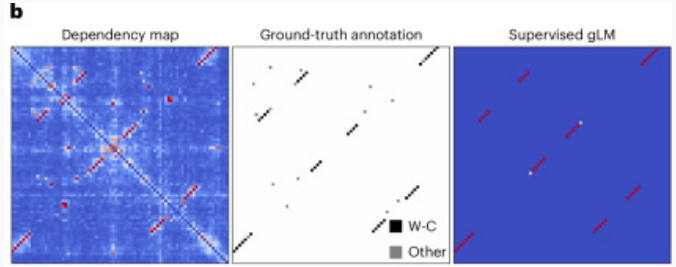
A centered 5×5 antidiagonal filter detects these patterns. Query substitutions favor Watson–Crick, and more weakly wobble, pairing.

RiNALMo dependencies predict contacts in Archive II: 3,865 RNAs across nine families; AUROC typically > 0.9 for most families.

RNAalifold and fine-tuned RiNALMo perform better on canonical secondary contacts.

Fig. 4a, cropped; Extended Data Fig. 4a–d; Methods. Archive II family sizes shown in panel.

Dependencies add contacts beyond secondary folds



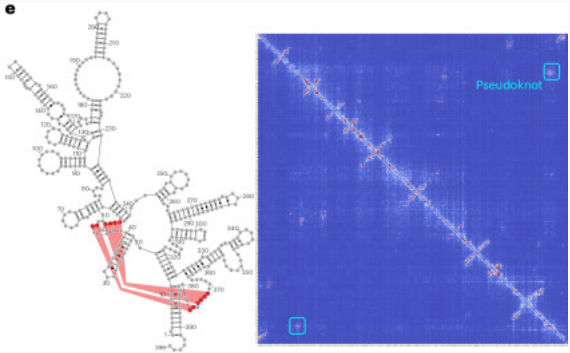
A. fulgidus tRNA: six of eight tertiary-only contacts have dependency > 6.

CompaRNA noncanonical contacts:
AUROC 0.793 (dependencies) versus 0.635 (supervised RiNALMo).

For pairs missed by the supervised model, dependency > 13.5 gives about 50% precision. This is a threshold-selected subset.

Fig. 4b,d, cropped; Fig. 4c; Methods. CompaRNA uses PDB contacts; canonical pairs excluded for this ROC.

Pseudoknots and competing folds survive in one map



RNase P: nonnested pseudoknot contacts form extra antiparallel stripes.

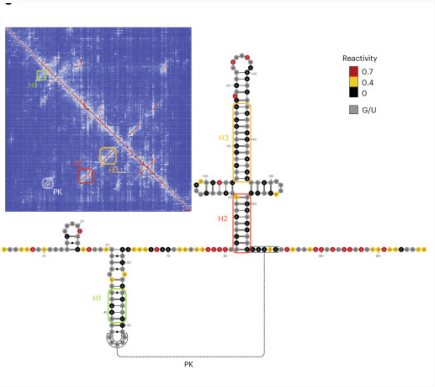
Pseudoknot benchmark
2,530 RNAs; 21,051 annotated pseudoknot contacts versus noncontact pairs. AUROC 0.92.

Tryptophan leader
Domain 3 depends on domains 2 and 4, consistent with alternative antiterminator and terminator folds.

The map can encode mutually exclusive possibilities; it does not measure their occupancies or kinetics.

Fig. 4e, cropped; Fig. 4f; Extended Data Fig. 4f. Similarity-filtered bpRNA-1m(90) dataset.

New RNA candidates: prediction followed by orthogonal evidence



Scan *E. coli* upstream noncoding regions, considering 100-, 200- and 500-base windows.

Select strong antiparallel patterns suggesting at least two stems; exclude annotated structures.

Four candidates upstream of *fkpB, glnS, mtlD, rlmB* are supported by in-cell DMS-MaPseq and covariation.

fkpB example: stem-loops H1–H3 plus a **putative** pseudoknot (PK).

Fig. 4g, cropped; Extended Data Fig. 4i; Methods, E. coli scan and covariation analysis.

What does the DMS experiment actually validate?

1. **Measurement**: in-cell DMS modifies accessible A/C bases; mutational profiling estimates reactivities. Low signal is consistent with protection, including base pairing.
2. **Genome-wide support**: bases in antiparallel dependencies have lower DMS mutation frequencies (5,008 A/C bases versus 103,867 others); pairs also receive higher DMS-constrained folding probabilities.
3. **Candidate support**: DMS-informed folds and sequence covariation corroborate four proposed structures.

Evidence boundary

DMS is primarily a per-base accessibility assay. DMS-constrained folding is a model-based inference of pairing; it does not directly measure every proposed pair or establish the regulatory function of the new structures.

Extended Data Fig. 4g–i; Methods, DMS-MaPseq and RNA covariation. Both genome-wide comparisons report $P < 10^{-16}$.

A stripe can also mean duplication

Forward repeats make parallel diagonals; **reverse-complement repeats** make antiparallel diagonals.

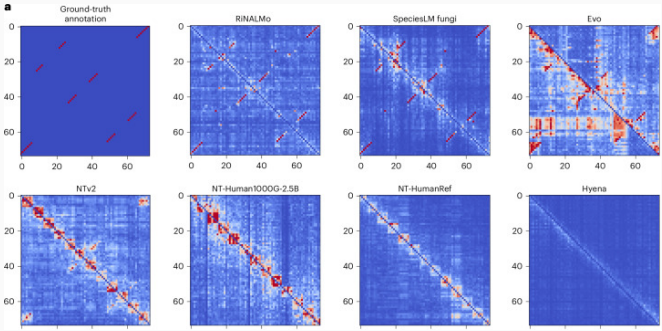
The authors insert new random repeats into synthetic backgrounds. Dependencies increase with repeat length, even when copies are far apart. This supports learning a duplication operation rather than memorizing those exact sequences.

At typical short tRNA stem lengths (2–7 bases), simulated reverse-complement repeats produce weaker dependencies than endogenous tRNA stems. The model therefore uses context beyond complementarity.

Interpretation
An antiparallel pattern alone is not sufficient evidence for a folded RNA. Ask whether the region is transcribed and whether probing, covariation or contact data support the structure.

Fig. 5; Extended Data Fig. 5; Supplementary Note, duplication experiments and matched tRNA controls.

Maps expose architecture artifacts and training limitations



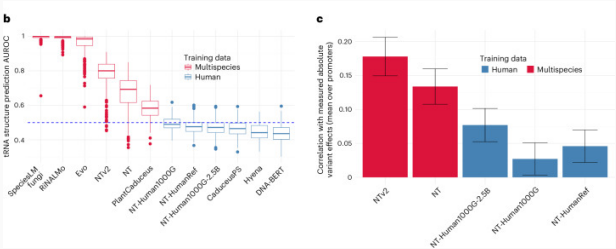
Six-mer tokens: artificial 6×6 diagonal blocks can masquerade as motifs.

Autoregression: one-sided context can miss a feature near its beginning; opposite strands partly mitigate this.

Model predictions and probing conventions jointly shape the map.

Fig. 7a, cropped. Same human tRNA, multiple models; colors visualize within-model dependencies.

Multispecies training matters more than human-only scale here



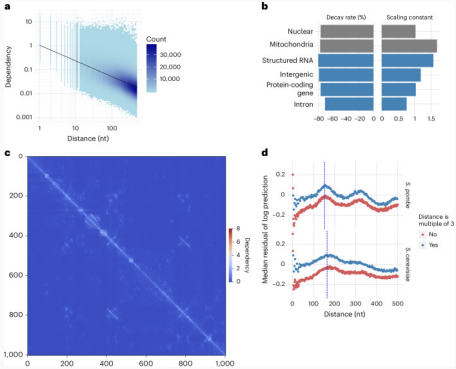
On 266 unique human tRNAs, multispecies models outperform human-only models, including models trained on diverse human genomes. Promoter variant prediction shows the same direction within the Nucleotide Transformer family.

Supported conclusion: human-only training misses these rare structured elements.

Scope: architectures and data differ; this is not a controlled universal scaling law.

Fig. 7b,c, cropped; Methods, model comparison. Boxplots show structure-level AUROC distributions.

Distance decay is both biology and a necessary null



Yeast dependencies decrease about 78% per tenfold increase in distance; mitochondrial dependencies are about 1.64× stronger.

Residual periodicities align with coding triplets and reported nucleosome spacings: 164 bp in *S. cerevisiae*, 152 bp in *S. pombe*.

These are pattern associations; the molecular origin of each peak is not established experimentally.

Any distal-element benchmark needs a distance-matched baseline.

Fig. 6, reproduced; Methods, genome-wide dependency distribution. Analyses restricted to separations at most 500 bp.

Model dependency is a hypothesis about biological coupling

What is measured	What needs validation
Prediction sensitivity	A causal variant effect in a living system
Strong pairwise entry	Direct contact versus shared motif, duplication or artifact
RNA contact enrichment	A unique 3D conformation, state occupancy, or a regulatory function
Benchmark performance	Generalization to new families, species and contexts

Absolute maxima erase direction. Extreme probabilities, unmasked targets, limited context and training biases affect interpretation.

A useful next experiment: disrupt each proposed partner and test compensatory rescue.

Discussion and Methods; interpretation and proposed experiment are journal-club analysis.

Discussion: a discovery tool with testable outputs

1. Which signal is most compelling: promoter variant effects, RNA contacts, or the discovery of four new structures? What independent test would change your confidence?
2. Would a dependency map add information after controlling for sequence identity, family membership and distance? How would you build a held-out-family benchmark?
3. For a candidate RNA stem, can double-mutant rescue separate pairing from a shared motif or indirect sequence context?

Take-home message
Perturbing a sequence model exposes structured relationships that reconstruction probabilities hide. The strongest evidence combines interpretable map patterns with task-specific controls and independent measurements.

Paper: doi:10.1038/s41588-025-02347-3. Code: github.com/gagneurlab/dependencies_DNALM; DMS data: GSE271937.